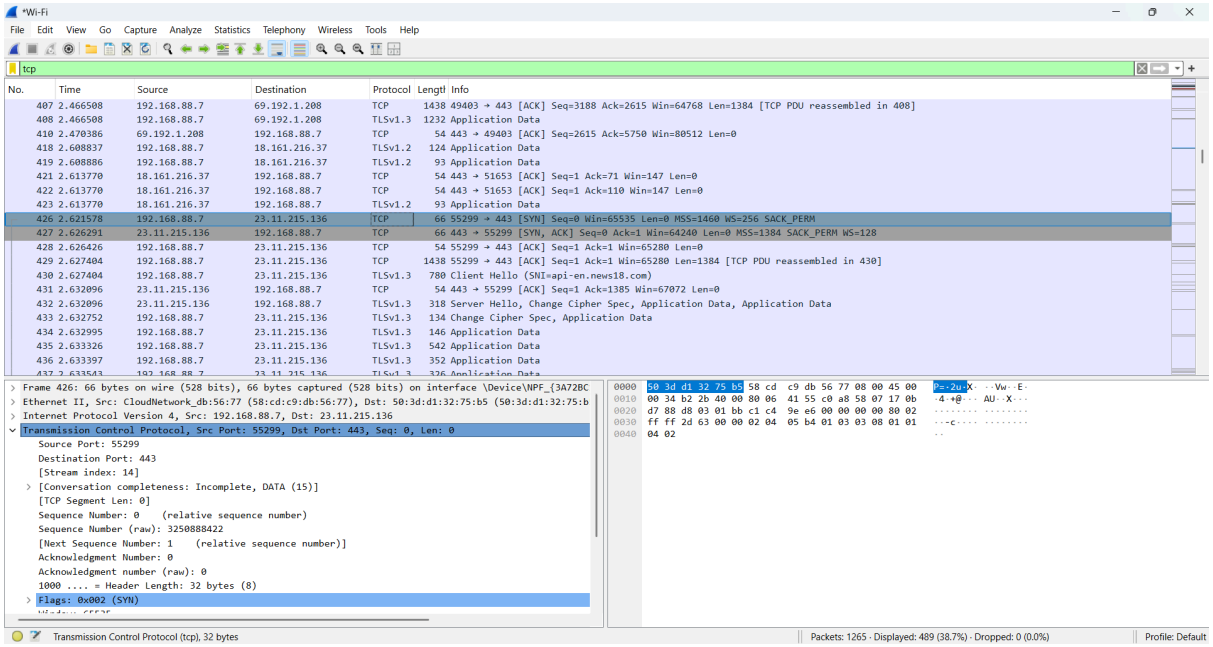
COMPUTER NETWORKS - UE24CS252B
Lab 4: TCP Congestion Window using Wireshark

Objective

The objective of this experiment is to study the mechanisms of TCP congestion control. The experiment aims to examine how the congestion window changes during data transmission, specifically identifying the conditions under which it grows and when it shrinks. It also seeks to understand which entity determines the size of the congestion window and how the client and server communicate this information during a TCP session. Additionally, the experiment investigates the circumstances that trigger the Fast Retransmit mechanism. Through this analysis, the experiment provides insight into how TCP manages network congestion and maintains reliable data transmission.

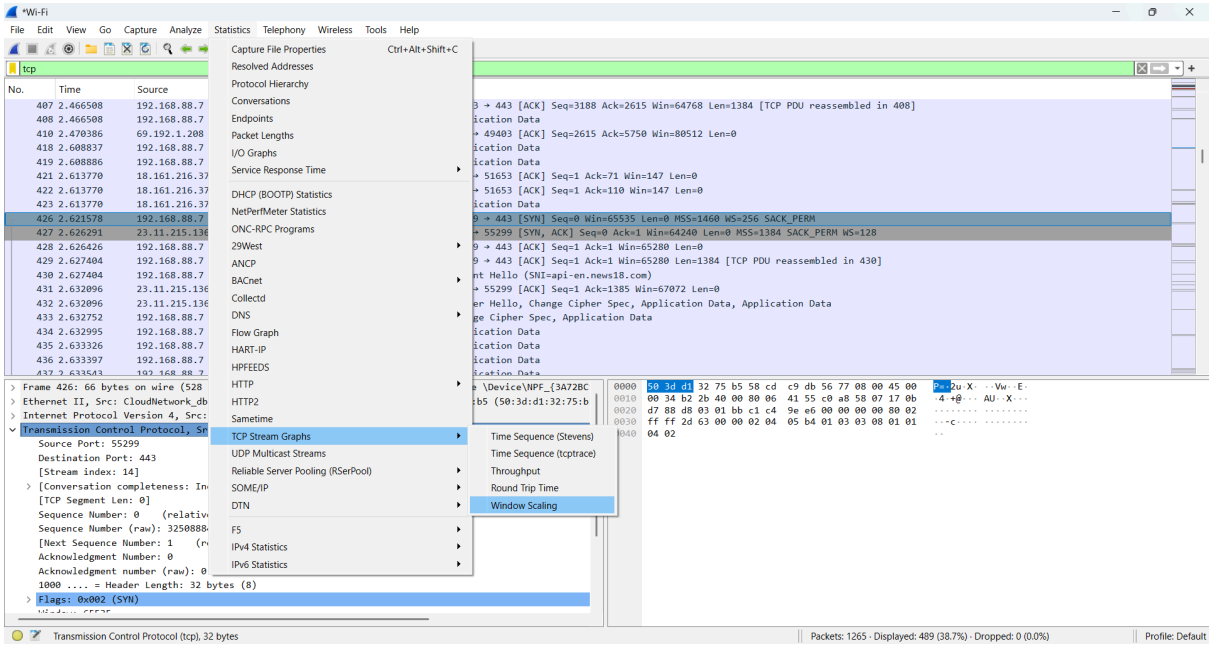
Task A: Capture TCP traffic and plot the Congestion Window:

> Open Wireshark in admin/sudo mode.
> Start the Capture on the live interface.
> Open a browser and visit a site of your choice (ex: www.news18.com)
> Apply the TCP filter in Wireshark

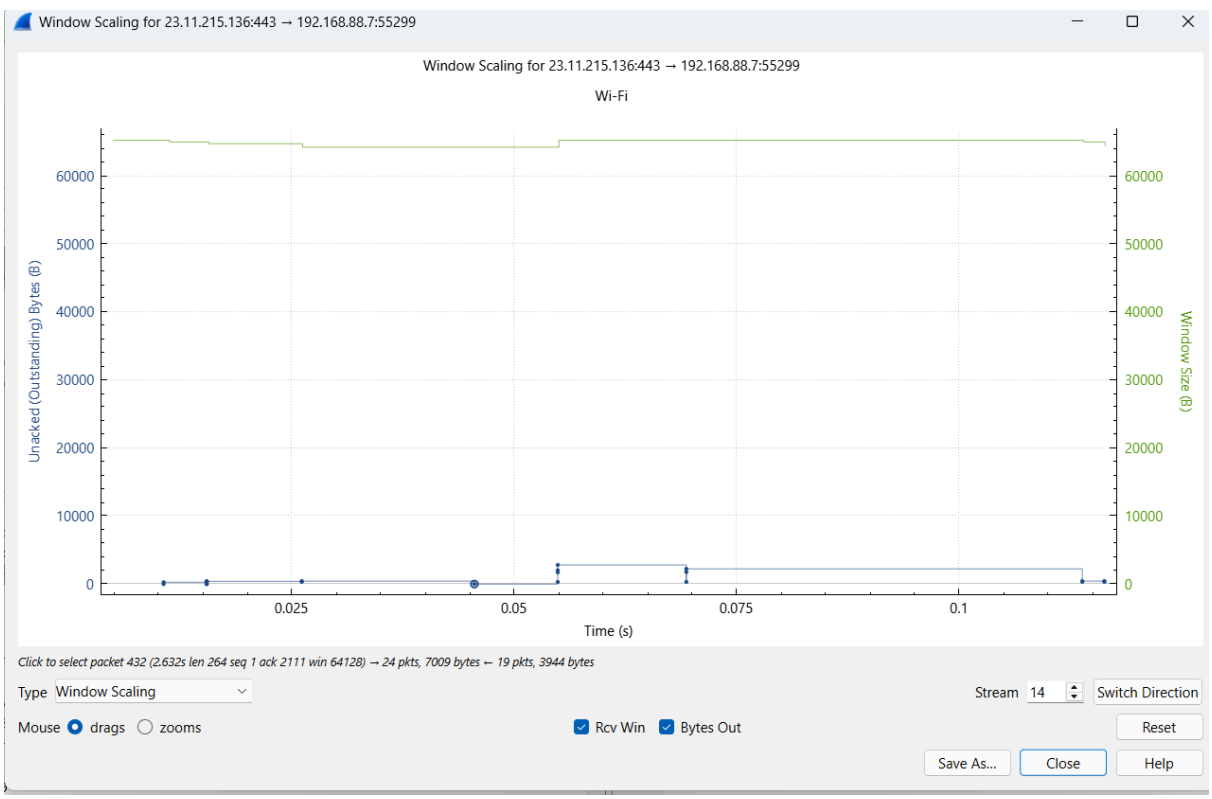


> Choose the very first TCP-SYN packet from your system to the server.
> Plot the Congestion Window:

> Statistics -> TCP Stream Graphs -> Window Scaling



> Look at the Congestion Window and the byte going out and coming in



Deliverables for Task A:

1. Screenshot of SYN , SYN-ACK, & ACK Packets from the Wireshark packet capture.

2. What is the sequence number of the TCP SYN segment used to initiate the TCP connection between the client and the server? Additionally, which field or flag in the TCP segment indicates that the segment is a SYN packet? [Paste the screenshot of the TCP segment details for the SYN packet.]

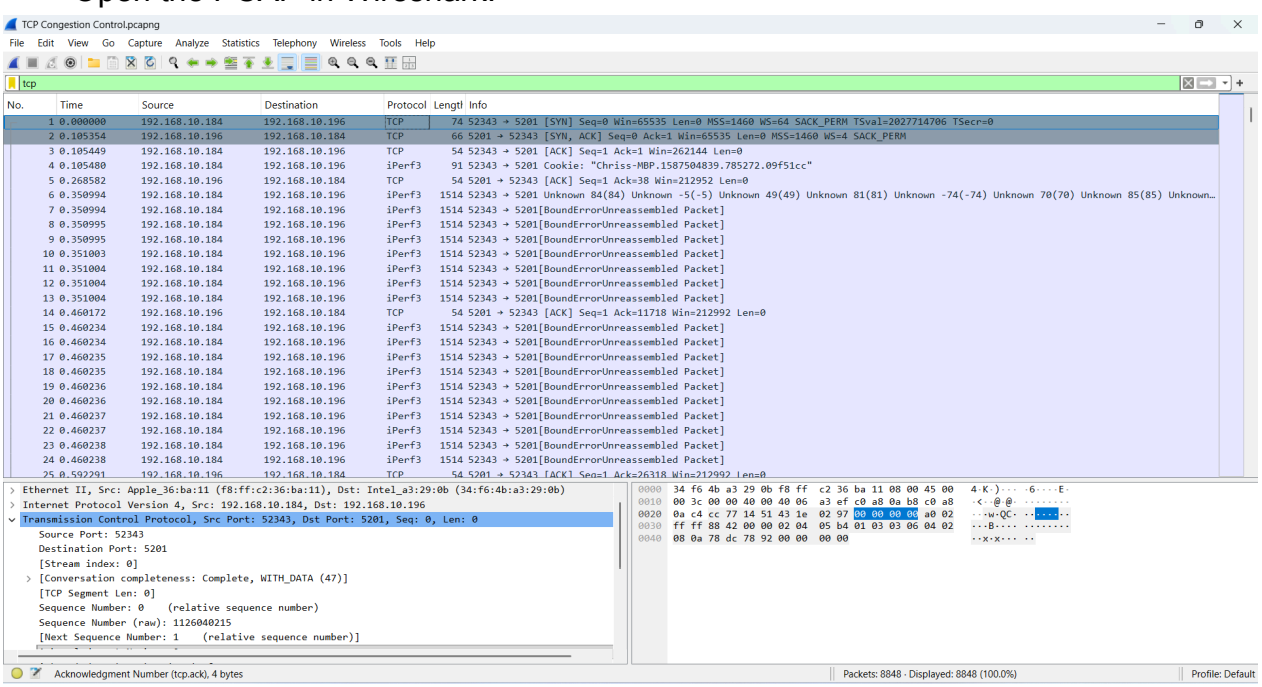
3. What is the sequence number of the SYN-ACK segment sent by the server to the client in response to the SYN request? Which field or flag in the TCP header indicates that the segment is a SYN-ACK packet? Additionally, what is the value of the Acknowledgement field in the SYN-ACK segment, and why is it set to that value?
   [Paste the screenshot of the TCP segment details for the SYN-ACK packet.]

4. Provide a screenshot of the congestion window observed in your experiment. What do the blue and green lines in the plot represent?

5. Based on the plot, what can you infer about TCP transmission behavior (indicate receiver-side or sender-side congestion control) and why?

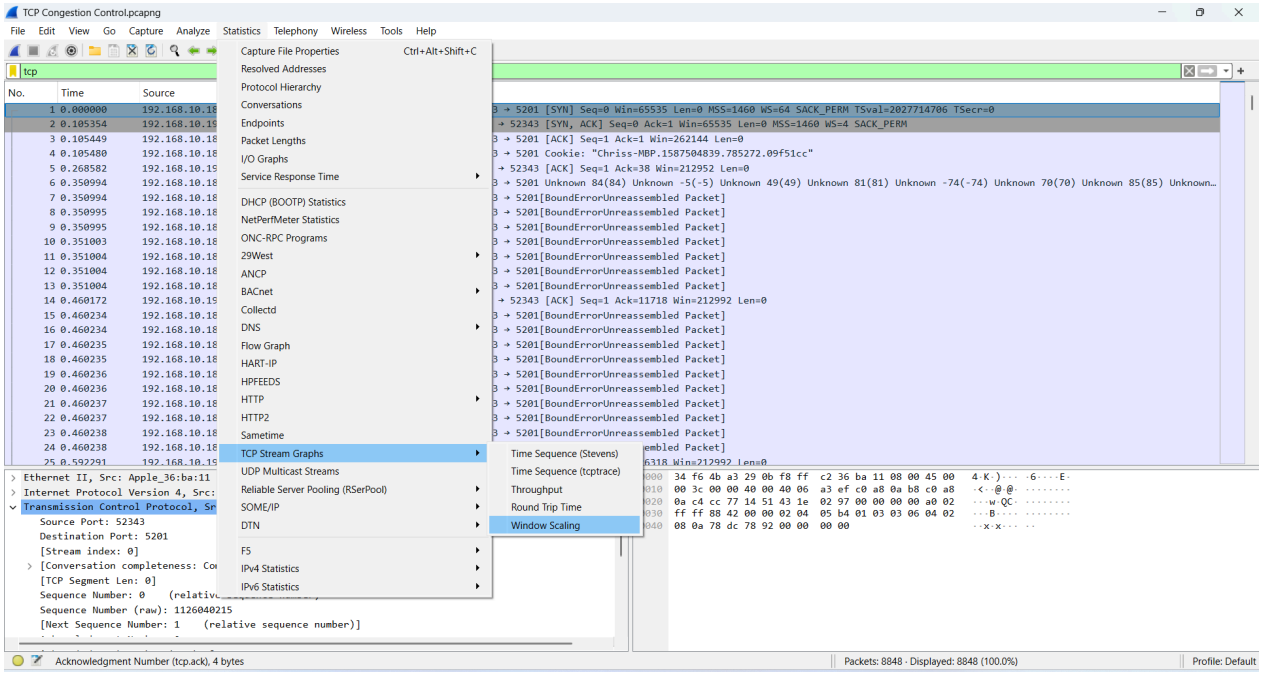
Task B: Analyzing the Congestion Window from the packet capture.

> Use the PCAP provided here to understand the TCP Congestion Control in case of 3 consecutive duplicate packets causing Fast Retransmit. (https://github.com/packetpioneer/youtube/blob/main/TCP%20Congestion%20Control.pcapng)
> Open the PCAP in Wireshark.

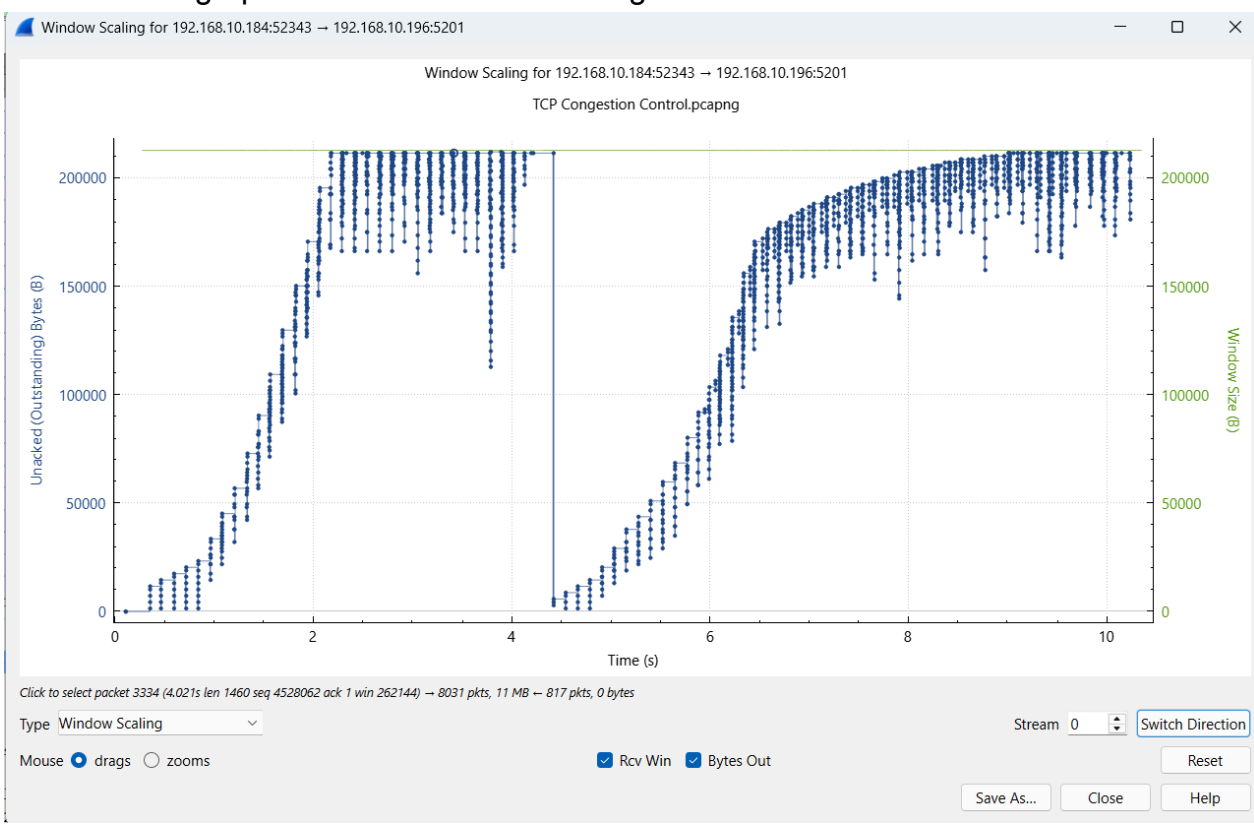


> Plot the Congestion Window through the same option as before

> Statistics -> TCP Stream Graphs -> Window Scaling



> Look at the graph and understand the Congestion Window. Switch the direction.



> Click on the fist dip in the graph to see the reason for it.
Using the congestion window find the packet at which there is dip.

Deliverables for Task B:

1. Provide a screenshot of the congestion window and clearly mark the different phases of TCP, such as Slow Start and Congestion Avoidance. Based on your observation of the graph, identify which TCP variant (flavor) is being used.

2. Examining the given PCAP file, provide a relevant screenshot that shows three duplicate ACKs in Wireshark. What do you observe in the subsequent packet transmission, and how would you explain this behavior?

3. Explain under what conditions the TCP congestion window (cwnd) decreases to 1, and describe the network event that causes this behavior.

4. Provide a screenshot of the packets observed in Wireshark after the congestion window is reduced to 1. Based on the subsequent packets, describe how the congestion window size changes over time.